Analysis of Mortality Rate by Education Level, Age Group, and Location Using Azure Databricks

Module: COM745 – Big Data and Infrastructure
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Problem & Background

- Educational attainment is widely recognised as a social determinant of health.
- Mortality rates often vary significantly across education levels, age groups, and geographic regions.
- Understanding these patterns at scale requires big data storage and distributed analytics.

Problem Statement:
To analyse how mortality rates vary by **education level**, **age group**, and **county-level location**, and how these patterns changed between **2018 and 2019**.

Dataset Selection & Relevance

- **Source:** Institute for Health Metrics and Evaluation (IHME) (https://ghdx.healthdata.org/record/ihme-data/us-le-county-education-2000-2019)
- **Geography:** United States (county level)
- **Years Used:** 2018 and 2019
- **Population Stratification:**
  - Education level
  - Age group
  - Location (county)

Why this dataset?

- Large-scale, longitudinal public health data
- Includes uncertainty bounds (`lower`, `upper`)
- Suitable for multi-dimensional analysis in a data lake

Analytical Objectives

- Examine the **education gradient** in mortality rates
- Compare changes between **2018 and 2019**
- Identify **age–education interactions**
- Highlight **geographic inequalities** in mortality outcomes
- Demonstrate use of **big data infrastructure** rather than traditional databases

Overall System Architecture

- Azure Databricks workspace deployed on Microsoft Azure
- Spark-based distributed processing
- Lakehouse architecture:
  - Bronze → Silver → Gold layers
- SQL + PySpark used for analytics and visualisation

Why Databricks?

- Scalability for large datasets
- Integrated analytics, storage, and visualisation
- Supports structured lakehouse workflows

Data Ingestion (Bronze Layer)

- Raw IHME CSV datasets for **2018** and **2019** ingested
- Files loaded into Databricks managed storage
- Schema inferred using Spark
- No transformation applied at this stage

Purpose: Preserve raw data for traceability and reproducibility

Data Cleaning & Transformation (Silver Layer)

Performed key cleaning operations:

- Removed rows with:
  - `edu_name = 'All educational attainment populations'`
  - Missing mortality values (`val IS NULL`)
- Converted mortality metrics to numeric types
- Retained uncertainty bounds (`lower`, `upper`)
- Dropped non-essential columns:
  - measure_id, location_id, sex_id, cause_id, metric_id, etc.

Outcome: Clean, analysis-ready dataset

Data Modelling (Gold Layer)

Created analytical aggregates:

- Mortality rate by:
  - Education level
  - Age group
  - Location
- Year-on-year comparison (2018 → 2019)
- County-level ranking of lowest mortality rates

Gold tables enabled:

- Fast querying
- Clear visualisation
- Insight-driven analysis

Scalability & Technology Justification

- Spark handles millions of records efficiently
- Queries execute in parallel across nodes
- Easily extensible to additional years or countries
- Alternative solutions (e.g., RDBMS) lack scalability and flexibility for this task

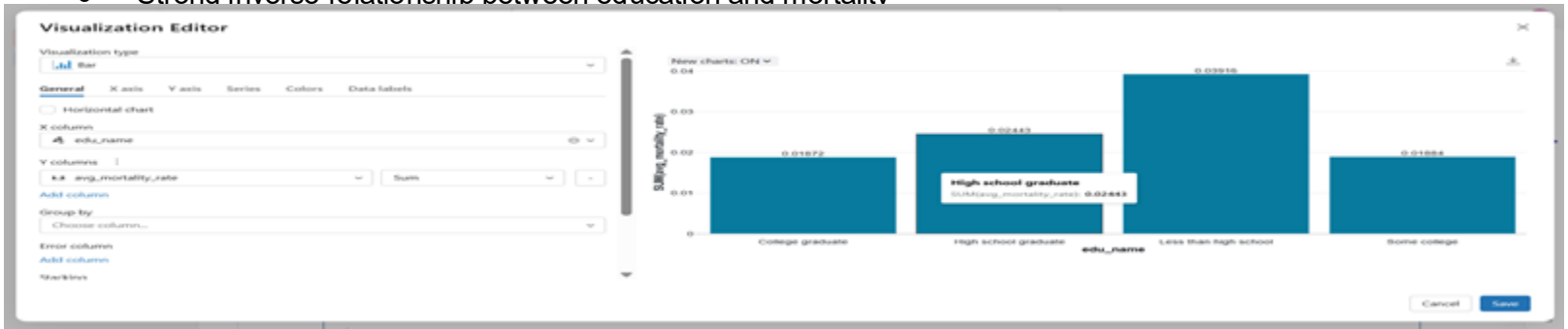
Insight 1: Education Gradient in Mortality

Key Observation:

- Mortality rates decrease consistently with higher educational attainment
- Individuals with **less than high school education** experience the highest mortality rates
- College graduates show the lowest mortality rates

Interpretation:

- Strong inverse relationship between education and mortality

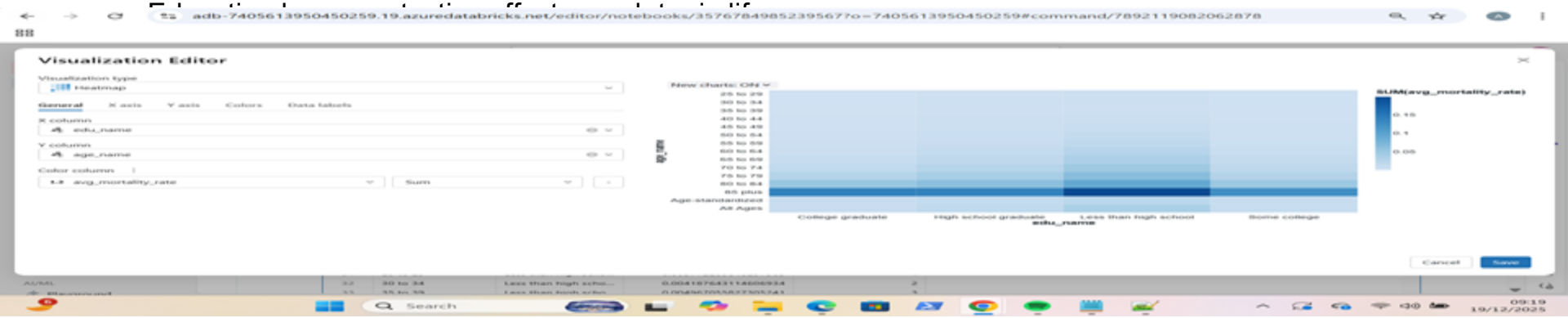


Insight 2: Age × Education Interaction

Key Observation:

- Mortality increases sharply with age across all education levels
- Education gradient persists at older ages
- Largest disparities observed in **75+ age groups**

Interpretation:



Insight 3: Geographic Inequality

Key Observation:

- Certain counties consistently show lower mortality rates
- Many low-mortality counties correspond to higher education levels
- Geographic clustering of favourable outcomes is evident

Interpretation:

- Location and education interact to shape health outcomes

Add filter

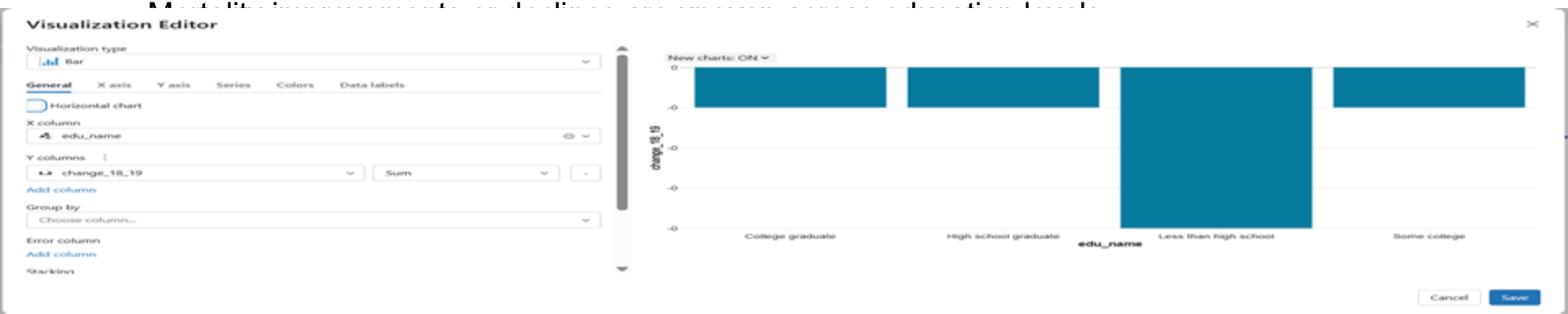
	location_name	edu_name	avg_mortality_rate
1	Summit County (Colorado)	College graduate	0.007239577581499138
2	Pitkin County (Colorado)	College graduate	0.007850099875853346
3	Eagle County (Colorado)	Some college	0.008780020712060374
4	Eagle County (Colorado)	College graduate	0.009173478895116731
5	Summit County (Colorado)	Some college	0.009309137512707986
6	Pitkin County (Colorado)	Some college	0.009598847791613496
7	Aleutians East Borough (Alaska)	Some college	0.009625320291281917
8	Aleutian Islands Census Area (Alaska)	Some college	0.009625320291281917
9	Aleutians West Census Area (Alaska)	Some college	0.009625320291281917
10	San Miguel County (Colorado)	College graduate	0.009755638203412699
11	Bethel Census Area (Alaska)	College graduate	0.009926121944852115
12	Snyder County (Pennsylvania)	College graduate	0.010357929012045073
13	North Slope Borough (Alaska)	Some college	0.010653295750100723
14	Sioux County (Iowa)	Some college	0.010759065894948 52
15	Wilbarger County (Texas)	Some college	0.010793582911743892
16	Hidalgo County (Texas)	College graduate	0.010847476079953 54
17	Juneau City and Borough (Alaska)	College graduate	0.010967391433041011
18	Prince George County (Virginia)	College graduate	0.011054802707343584
19	Stephenson County (Illinois)	Some college	0.011087178226263683
20	Cook County (Minnesota)	College graduate	0.011139640248416351
21	Braxton County (West Virginia)	Some college	0.011149876779203911
22	Maui County (Hawaii)	College graduate	0.011196473167722498
23	Kalawao County (Hawaii)	College graduate	0.011196473167722498
24	McIntosh County (Georgia)	Some college	0.011202773593292449
25	Grand County (Colorado)	College graduate	0.011243399969641771

Insight 4: Change Between 2018 and 2019

Key Observation:

- All education groups show slight changes in mortality rates
- The largest magnitude of change occurs in the **lowest education group**
- Changes are relatively small but consistent

Interpretation:



Video Demonstration

Concluding Comments

- Educational attainment shows a strong and persistent relationship with mortality
- Age amplifies existing education-based disparities
- Geographic location further influences mortality outcomes
- Azure Databricks proved effective for scalable health data analysis

Limitations & Future Work:

- Extend analysis to additional years
- Include socioeconomic variables such as income
- Apply predictive modelling for future mortality trends

References

[1] Institute for Health Metrics and Evaluation (IHME), *United States Mortality by Education and County, 2000–2019*, Public Use Dataset, 2023. [Online]. Available: https://ghdx.healthdata.org

[2] Microsoft, *Azure Databricks Documentation*, Microsoft Learn, 2025. [Online]. Available: https://learn.microsoft.com/azure/databricks

[3] Apache Software Foundation, *Apache Spark SQL, DataFrames and Datasets Guide*, 2025. [Online]. Available: https://spark.apache.org/sql

[4] M. Marmot et al., “Social determinants of health inequalities,” *The Lancet*, vol. 365, no. 9464, pp. 1099–1104, 2005, doi: 10.1016/S0140-6736(05)71146-6.

[5] World Health Organization, *Social Determinants of Health*, WHO, 2023. [Online]. Available: https://www.who.int/teams/social-determinants-of-health